

Attention-Enhanced Deep Learning for Multi-Class Brain Tumor MRI Classification

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Abstract—This paper presents an end-to-end deep learning approach for multi-class brain tumor image classification from MRI images. The presented approach solves problems encountered during the analysis process of brain MRI images using preprocessing, data augmentation, and transfer learning or neural networks. These images represented 44 classes, consisting of diverse species related to their images from T1, T1+C+, T2 sequences, and normal images of human brains. Varieties of CNN architectures have also been applied, including ResNet-18, DenseNet-121, VGG-16, CNN, and attention CNN, where accuracy, precision, recall, F1-score, ROC AUC, execution time, and explanation plots represent respective GradCAM outputs. It is observed that accuracy is maximal for DenseNet-121, and accuracy and complexity possess the best tradeoff for ResNet-18.

Index Terms—Brain Tumor, MRI, Deep Learning, Transfer Learning, Attention Mechanism, Grad-CAM

I. INTRODUCTION

Diagnosis of a brain tumor by an MRI machine is a complex process in terms of the anatomical structure of the brain, variations depending on modalities, and overlapping areas in different types of tumors. It takes a longer time to diagnose by manual analysis by the radiologists, resulting in the use of automated systems based on deep learning techniques.

In specific, this work aims to cover fine-grained multi-class classification problems where tumors will be categorized not only according to their nature but also according to their MRI image sequences. In addition to this, a complete processing pipeline is also processed from raw MRI data to explainable deep learning models.

II. DATASET DESCRIPTION

A total of 44 classes, all depicting brain MRI scans, make up the dataset. These classes include specific varieties of tumors and scans such as Astrocytoma T1, Astrocytoma T1C+, Astrocytoma T2, and other normal scans of the brain.

The dataset exhibits:

- Class imbalance across tumor types
- Variation in image resolution
- Presence of non-brain regions such as skull and background

- Noise and contrast variability inherent to MRI scans

An initial exploratory analysis confirms the diversity of image resolutions and class distributions, motivating a structured preprocessing strategy.

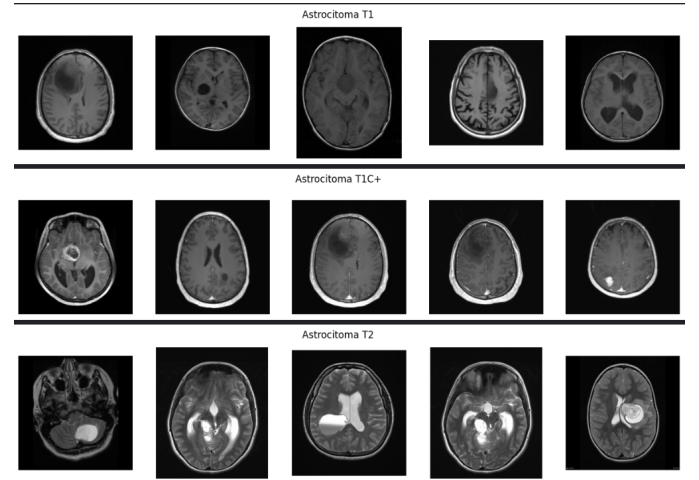


Fig. 1. Dataset folder structure and class distribution output.

III. PREPROCESSING PIPELINE

A multi-stage preprocessing pipeline is designed to standardize image quality, enhance discriminative features, and remove irrelevant anatomical structures.

A. Skull Stripping

Skull area and background area are eliminated with the help of a thresholding method-based skull stripping technique. Otsu's thresholding method effectively generates a binary image out of which the largest connected area with the brain area can be obtained using a connected component analysis method in a technical aspect related to the examination of the binary image.

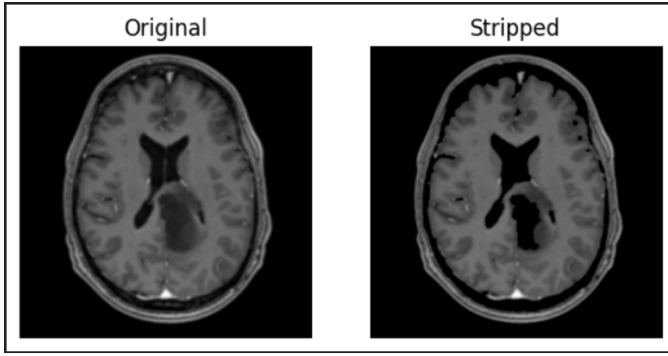


Fig. 2. Skull stripping: original image vs masked brain image.

B. Brain Region Cropping and Resizing

Next, the brain region is identified within the skull strip region based on the non-zero mask area. All images are then resized to a resolution of 224×224 pixels to be compatible with deep learning models.

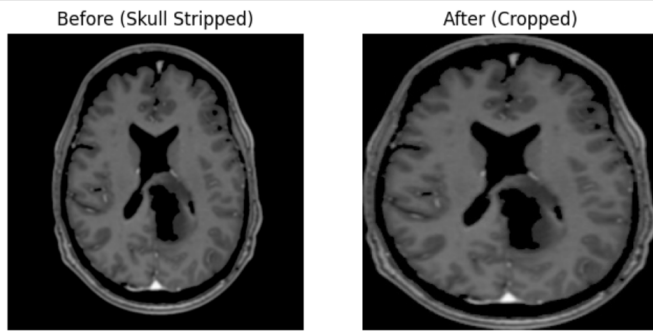


Fig. 3. Cropping process: before and after cropping.

C. Image Denoising

To suppress MRI noise while preserving structural information, non-local means denoising is applied to all cropped images. This step improves signal clarity and reduces high-frequency noise artifacts.

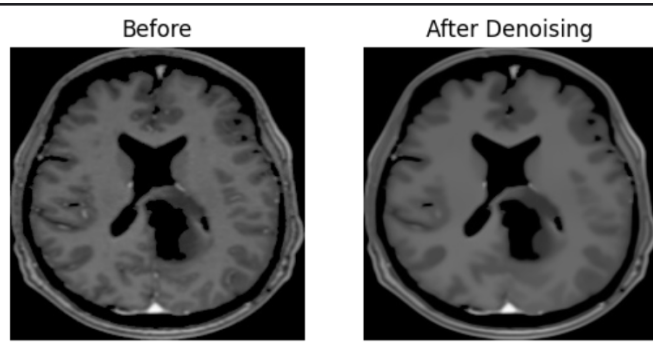


Fig. 4. Denoising: before and after comparison.

D. Contrast Enhancement using CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) is employed to enhance local contrast and highlight subtle tumor regions. This technique improves visibility of tumor boundaries without amplifying noise.

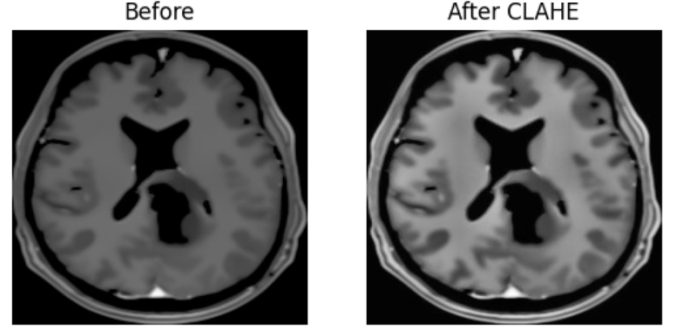


Fig. 5. CLAHE enhancement: before and after comparison.

E. Preprocessing Summary Visualization

A comprehensive before-and-after visualization is generated, showing original images alongside fully preprocessed outputs. This confirms the effectiveness of skull removal, cropping, denoising, and contrast enhancement.

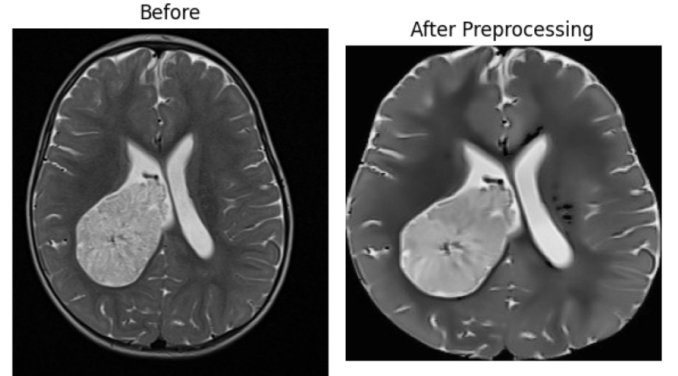


Fig. 6. Preprocessing summary grid (original vs final processed images).

IV. DATA AUGMENTATION AND DATASET SPLITTING

A. Data Augmentation

To address class imbalance and improve generalization, data augmentation is applied to the preprocessed images. Augmentation techniques include small rotations, translations, zooming, and brightness adjustments. Each original image is augmented multiple times, expanding the dataset size while preserving anatomical realism.

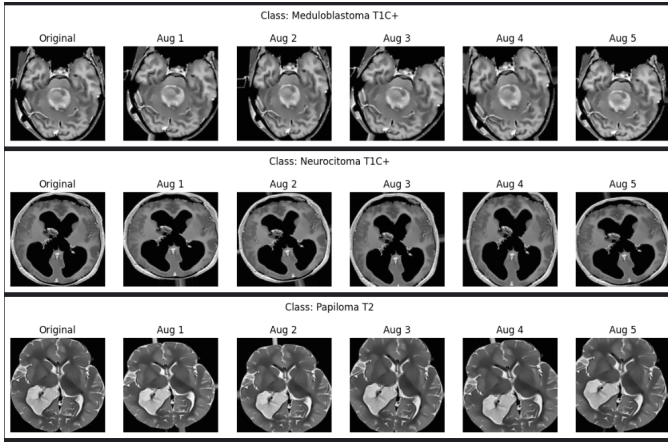


Fig. 7. Data augmentation examples (original and augmented images).

B. Train-Validation-Test Split

The augmented dataset is divided into training, validation, and test sets using a 70% / 15% / 15% split. The final split results in:

- Training set: 15,653 images
- Validation set: 3,344 images
- Test set: 3,390 images

V. MODEL ARCHITECTURES AND TRAINING STRATEGY

Five neural network models are evaluated:

- 1) ResNet-18 (transfer learning)
- 2) DenseNet-121 (transfer learning)
- 3) VGG-16 (transfer learning)
- 4) Custom CNN (trained from scratch)
- 5) Attention-enhanced CNN with Squeeze-and-Excitation blocks

Transfer learning models leverage ImageNet-pretrained weights, while custom models are trained from scratch. Early stopping and learning rate scheduling are used to prevent overfitting. All models are trained and evaluated under identical data splits for fair comparison.

VI. MODEL IMPLEMENTATION DETAILS

This section summarizes what was performed for each model in the notebook workflow. For all models, the same train/validation/test split was used, and evaluation was performed using accuracy, precision, recall, F1-score, ROC-AUC, and runtime (training and inference time).

A. ResNet-18 (Transfer Learning)

For the ResNet-18, the weights were initialized with Imagenet, and only the final classification layer was replaced, allowing the network to be adjusted for the 44-class task. The MRI images were preprocessed by resizing them to 224x224 pixels, then Normalized using statistics for the Imagenet dataset, according to what the network was pretrained on. For optimization, mini-batch optimization was utilized, with the goal of validating on the network. After completion, the trained

network was then evaluated on the test set, with a confusion matrix, ROC curves, and other classification results provided.

B. DenseNet-121 (Transfer Learning)

DenseNet-121 was also initialized with ImageNet weights and adapted for a 44-class output at the end of the classifier. Preprocessing and normalizing inputs remained the same. The training procedure also remained the same and was based on optimization per epoch and early stopping based on the validation performance. Finally, for convergence analysis, evaluation was conducted based on test accuracy, macro and micro ROC-AUC, confusion matrices, and class-wise performance. DenseNet-121 was chosen based on performance and utilized for the explanation analysis in the final stages using Grad-CAM.

C. VGG-16 (Transfer Learning)

VGG-16 was used as a transfer learning baseline by initializing pretrained weights and replacing the final fully connected classification layer for 44 categories. Images were resized to match the expected input shape. The model was trained under the same split and evaluated using the same metrics. The final reported test accuracy is included in the results section.

D. Custom CNN (From Scratch)

A custom convolutional neural network (CNN) was designed and trained from scratch without pretrained weights. The architecture learned features directly from the MRI dataset. Training was carried out using the same split and the same evaluation protocol. After training, test performance was measured and compared against transfer learning models to highlight the advantage of pretrained backbones in multi-class MRI classification.

E. Attention-Enhanced CNN (Squeeze-and-Excitation)

To improve the baseline CNN, an attention-enhanced CNN was implemented by introducing Squeeze-and-Excitation (SE) blocks for channel-wise feature recalibration. This allows the network to emphasize informative feature maps and suppress less relevant ones. The attention model was trained and evaluated using the same split and metric pipeline. Its performance was compared against the standard custom CNN to quantify the benefit of the attention mechanism.

VII. EXPERIMENTAL RESULTS

A. ResNet-18 Results

ResNet-18 achieved strong performance with a favorable efficiency profile:

Test Accuracy: 98.82%

Macro AUC: 0.9999

Micro AUC: 1.0000

Average inference time: 5.55 ms per image

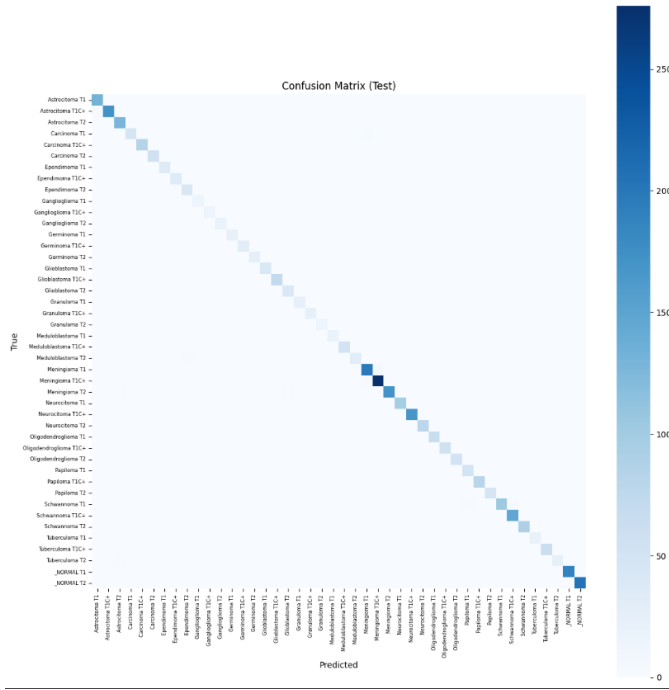


Fig. 8. ResNet-18 confusion matrix.

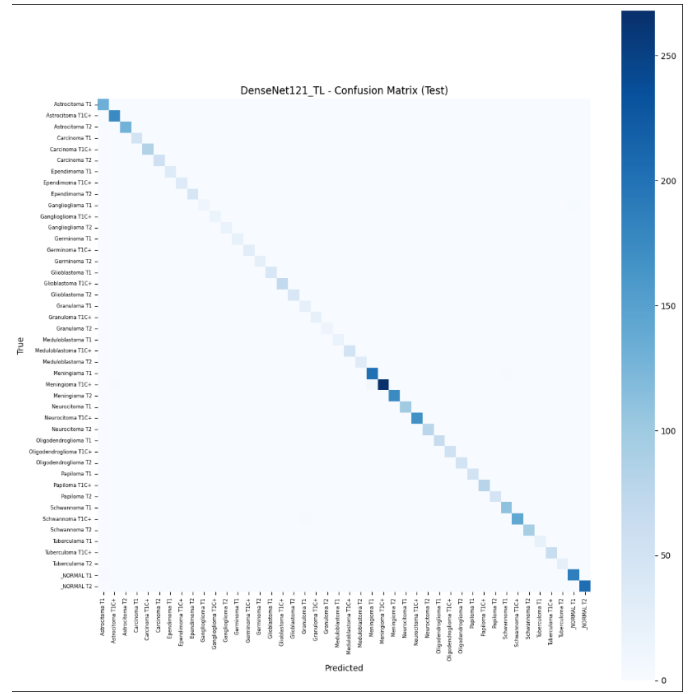


Fig. 10. DenseNet-121 confusion matrix.

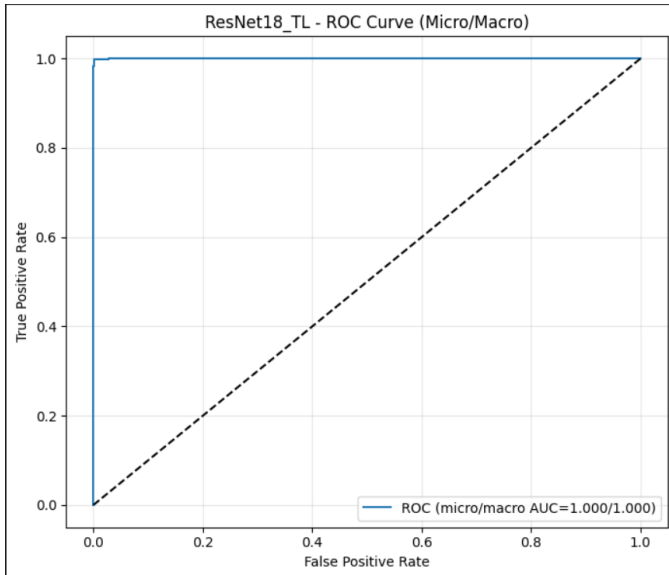


Fig. 9. ResNet-18 ROC curves.

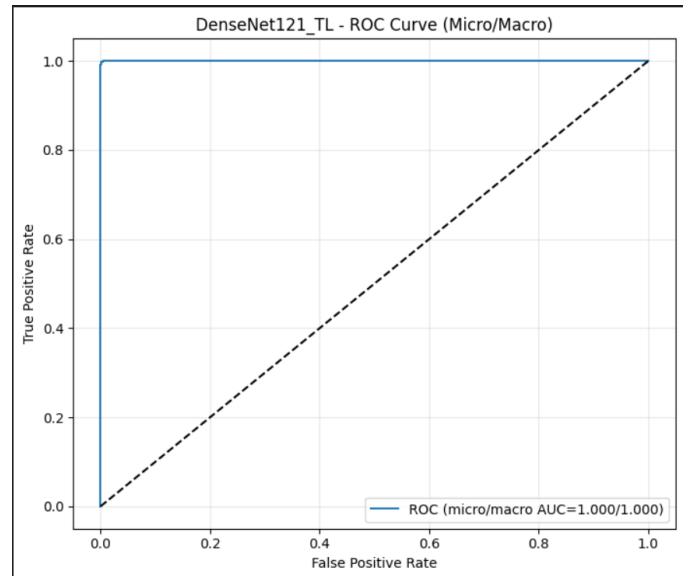


Fig. 11. DenseNet-121 ROC curves.

B. DenseNet-121 Results

DenseNet-121 achieved the highest overall accuracy but required higher inference time:

Test Accuracy: 99.17%

Macro AUC: 1.0000

Micro AUC: 1.0000

Average inference time: 19.00 ms per image

C. VGG-16 Results

VGG-16 achieved strong classification performance but with higher computational cost than ResNet-18.

Test Accuracy: 97.49%

(Detailed evaluation plots were not retained in the notebook output.)

D. Custom CNN Results

The custom CNN trained from scratch produced lower accuracy than pretrained models but remained efficient:

Test Accuracy: 95.10%

Macro AUC: 0.9996

Average inference time: 4.58 ms per image

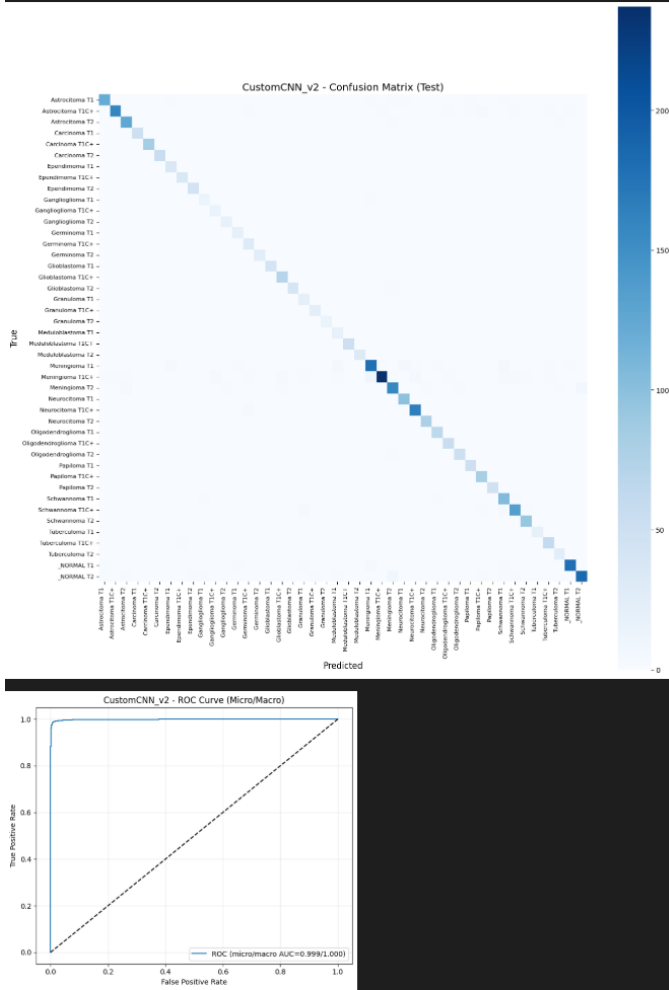


Fig. 12. Custom CNN confusion matrix and ROC curves.

E. Attention-Enhanced CNN Results

The attention-enhanced CNN improved over the baseline CNN by emphasizing informative features:

Test Accuracy: 96.70%

Macro AUC: 0.9998

Average inference time: 5.41 ms per image

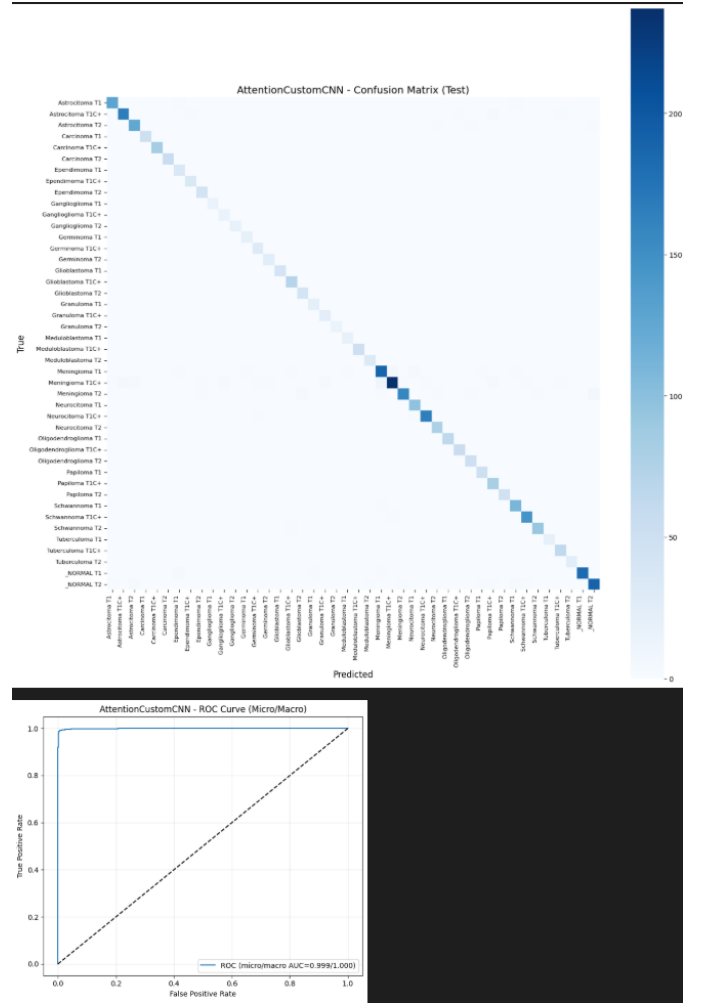


Fig. 13. Attention CNN confusion matrix and ROC curves.

VIII. GRAD-CAM EXPLAINABILITY

To interpret the decision-making process of the best-performing model (DenseNet-121), Gradient-weighted Class Activation Mapping (Grad-CAM) was applied. Grad-CAM produces heatmaps that indicate the spatial regions in an MRI image that contribute most strongly to a predicted class.

In this work, Grad-CAM visualizations were generated for two scenarios: (i) correctly classified samples across the dataset and (ii) misclassified samples where the model prediction differed from the ground truth. These visualizations help verify whether the model focuses on clinically meaningful tumor regions rather than irrelevant background patterns.

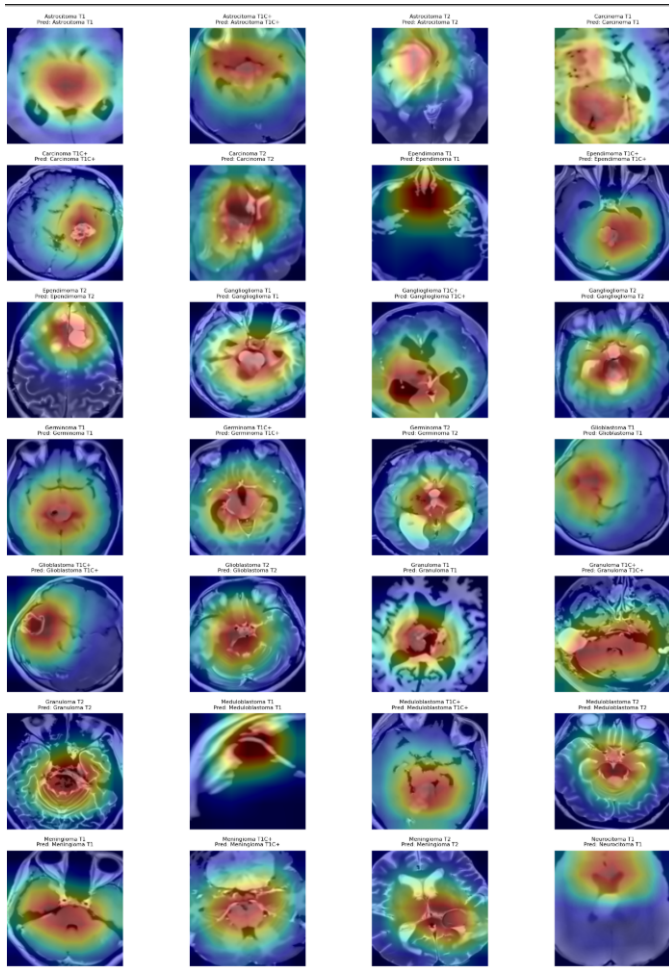


Fig. 14. Grad-CAM examples for correct predictions (insert sample overlays).

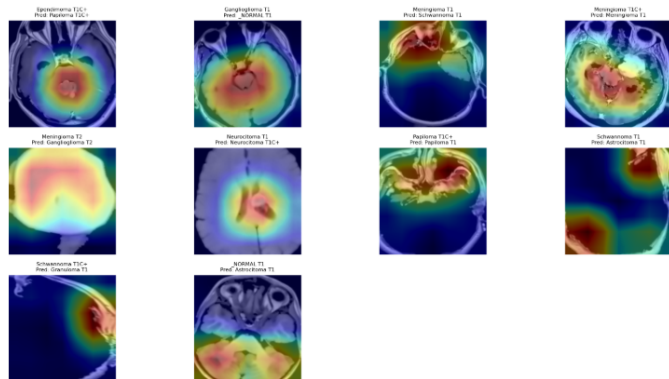


Fig. 15. Grad-CAM examples for incorrect predictions (insert sample overlays).

IX. DISCUSSION

The results show that there is a substantial accuracy gain for transfer learning over training a network from scratch in multi-class brain tumor image classification. DenseNet-121